

LEVEL 4 – Very Tough HC Verma / JEE Advanced Mains-level Questions (Matrices)

1. Determinants & Advanced Properties

1. Evaluate $\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$.
 2. Find the determinant of a 4×4 matrix using block matrices.
 3. Show that $\det(A + B) \neq \det(A) + \det(B)$ with a 3×3 counterexample.
 4. Compute $\det \begin{bmatrix} 1 & x & x^2 & x^3 \\ 1 & y & y^2 & y^3 \\ 1 & z & z^2 & z^3 \\ 1 & w & w^2 & w^3 \end{bmatrix}$ (Vandermonde determinant).
 5. If A is skew-symmetric, prove that $\det(A) = 0$ for odd n .
 6. Find determinant of matrix A whose rows are in arithmetic progression.
 7. Evaluate $\det \begin{bmatrix} a & b & c & d \\ b & c & d & a \\ c & d & a & b \\ d & a & b & c \end{bmatrix}$ (circulant matrix).
 8. If two rows of a determinant are identical, prove it is zero.
 9. Use cofactor expansion to compute 5×5 determinant with many zeros.
 10. Determine determinant after one row multiplied by scalar k .
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2. Eigenvalues, Eigenvectors & Diagonalization

11. Diagonalize $A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}$.
 12. Find eigenvalues of 3×3 matrix with row sums equal.
 13. If $A^2 = A$, find eigenvalues of A .
 14. Show that eigenvectors of symmetric matrix corresponding to distinct eigenvalues are orthogonal.
 15. If λ is eigenvalue of A , show λ^n is eigenvalue of A^n .
 16. Find eigenvalues of skew-symmetric 3×3 matrix.
 17. Compute eigenvectors of 2×2 rotation matrix.
 18. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find P such that $P^{-1}AP$ is diagonal.
 19. Prove Cayley-Hamilton theorem for 3×3 matrix.
 20. Verify that sum of eigenvalues = trace(A), product = determinant(A) for 3×3 matrix.
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3. Inverse, Adjoint & Advanced Systems

21. Solve system using inverse of 3×3 matrix with fractions.
 22. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, find A^{-1} using elementary operations.
 23. Prove $(A^{-1})^{-1} = A$.
 24. Solve system $AX = B$, where A has parametric entries.
 25. If $A^2 - 5A + 6I = 0$, express A^{-1} as a linear combination of I and A .
 26. Find inverse of 4×4 upper triangular matrix.
 27. Solve 4×4 system using block matrix inversion.
 28. Prove that product of invertible matrices is invertible, find inverse.
 29. Find inverse of nilpotent 3×3 matrix, if possible.
 30. Compute A^{-1} for diagonalizable 3×3 matrix using eigenvectors.
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4. Rank & Advanced Linear Systems

31. Determine rank of 5×5 matrix with rank deficiency 1.
 32. Solve system of 4×4 linear equations using rank method.
 33. Determine rank of triangular 4×4 matrix with one zero diagonal.
 34. Find rank of skew-symmetric 4×4 matrix.
 35. System: $\text{rank}(A) = 2$, $\text{rank}([A|B]) = 2$, $n = 3 \rightarrow$ find general solution.
 36. Solve 3×3 parametric system and find values of parameter for unique solution.
 37. Determine nullity of 4×4 matrix using rank-nullity theorem.
 38. Show that $\text{rank}(A) = \text{rank}(A^T)$ using 3×3 example.
 39. Determine rank of matrix after row operations with parameters.
 40. Solve linear system with infinite solutions and express general solution.
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5. Special Matrices (Orthogonal, Idempotent, Nilpotent)

41. Find a 3×3 orthogonal matrix with determinant -1 .
 42. Find 2×2 nilpotent matrix, verify eigenvalues $= 0$.
 43. Construct 2×2 idempotent matrix, verify $A^2 = A$.
 44. Show that product of orthogonal matrices is orthogonal.
 45. Find eigenvalues of idempotent 3×3 matrix.
 46. Prove that trace of nilpotent matrix $= 0$.
 47. Determine if sum of two idempotent matrices is idempotent.
 48. Construct 3×3 orthogonal matrix with complex entries.
 49. Find inverse of orthogonal matrix using transpose.
 50. Verify property: $\text{tr}(AB) = \text{tr}(BA)$ for 3×3 matrices.
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6. Matrix Transformations & Geometry

51. Rotate triangle vertices by 90° about origin using matrix.
 52. Reflect point across $y = x$ line using matrix multiplication.
 53. Scale rectangle: 2 along x, 3 along y-axis, find new coordinates.
 54. Combine rotation and scaling: find transformed coordinates.
 55. Use matrix to compute area of transformed triangle.
 56. Apply rotation + reflection to a point (1,2).
 57. Transform vector (x,y,z) using 3×3 scaling matrix.
 58. Use 2×2 matrix to rotate vector by θ .
 59. Find matrix representing reflection across y-axis.
 60. Use Markov chain 2×2 transition matrix to find steady state vector.
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7. Advanced Mixed Problems

61. Solve system using Cramer's rule with 4×4 determinant.
 62. Find values of parameter for which system has unique solution.
 63. Solve 3×3 system using inverse method with fractions.
 64. Evaluate 4×4 determinant using properties (zeros, row operations).
 65. Verify Cayley-Hamilton theorem for 4×4 matrix.
 66. Diagonalize 3×3 symmetric matrix.
 67. Find sum of eigenvalues of A^3 given A eigenvalues.
 68. Prove that determinant of orthogonal matrix = ± 1 .
 69. Show that nilpotent matrix has only zero eigenvalues.
 70. Solve system with singular 3×3 matrix and find general solution.
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8. Very Tough Conceptual MCQs

71. Construct 3×3 matrix with eigenvalues 1,2,3.
 72. Find inverse using adjoint for 3×3 matrix with fractions.
 73. Prove that product of idempotent matrices is not necessarily idempotent.
 74. Determine all possible eigenvalues of 2×2 orthogonal matrix.
 75. Show that determinant of skew-symmetric matrix of odd order = 0.
 76. Solve system where one equation is linear combination of others.
 77. Use rank to determine solution type of parametric system.
 78. Compute trace of square of 3×3 matrix using eigenvalues.
 79. Diagonalize rotation matrix 2×2 .
 80. Find matrix that scales x by 2, rotates 90° CCW, reflects across y-axis.
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9. Applications & Geometry Advanced

81. Use matrix method to find area of parallelogram in 3D.

82. Transform cube vertices using 3×3 scaling + rotation matrix.
 83. Markov chain: 3×3 transition matrix, find steady state.
 84. Compute eigenvalues of transition matrix.
 85. Apply sequence of transformations to point and find final coordinates.
 86. Use matrix to rotate triangle and compute area of rotated triangle.
 87. Use matrix to scale and reflect polygon.
 88. Compute volume of parallelepiped using determinant.
 89. Verify determinant property for transformed cube.
 90. Find coordinates after combined transformations using matrix multiplication.
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10. Challenge Multi-step Problems

91. Solve 4×4 system with parameter, find values giving unique solution.
 92. Compute inverse of block-diagonal matrix.
 93. Verify Cayley-Hamilton for 3×3 matrix with fractions.
 94. Construct 3×3 nilpotent matrix and verify.
 95. Diagonalize symmetric 4×4 matrix.
 96. Solve system of 3×3 equations using inverse and verify solution.
 97. Compute determinant of matrix with all entries 1 except diagonal 0.
 98. Find eigenvectors of 3×3 matrix with repeated eigenvalues.
 99. Determine rank of 4×4 matrix using minors.
 100. Solve parametric 3×3 system and express general solution.
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11. Extra Very Tough Advanced Problems

101. Construct 3×3 matrix with given eigenvalues and trace.
102. Determine if 3×3 matrix is diagonalizable.
103. Use eigenvalues to compute powers of 3×3 matrix.
104. Solve 3×3 system with singular coefficient matrix.
105. Verify $\text{tr}(AB) = \text{tr}(BA)$ for given 3×3 matrices.
106. Find matrix representing combined rotation and reflection in 2D.
107. Solve 4×4 system with fractions using Cramer's rule.
108. Find rank of matrix with rows linear combinations of others.
109. Compute determinant using row operations for 5×5 matrix.
110. Solve system with rank-deficient 3×3 matrix.